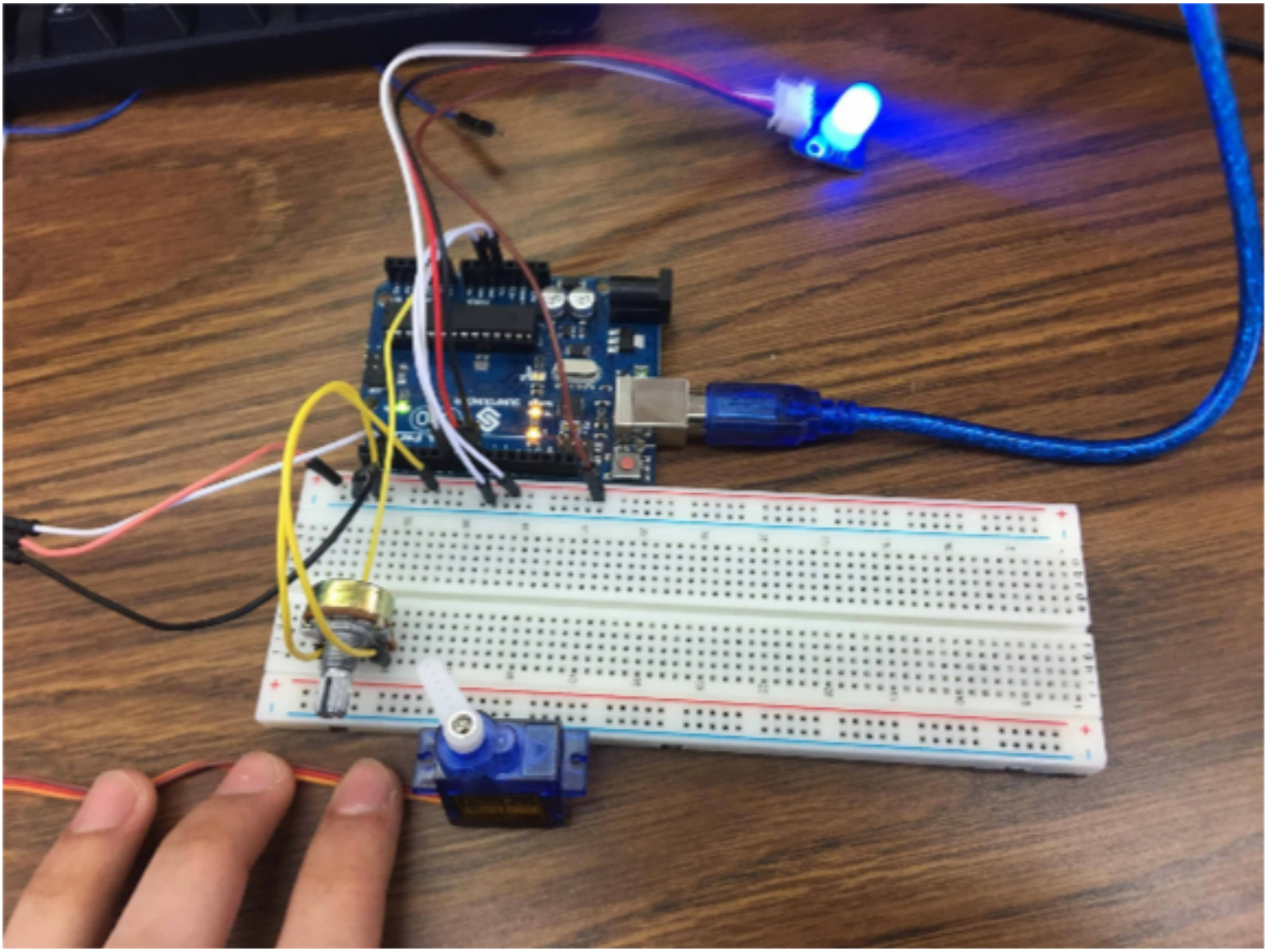

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```
//TITLE:ACTIVITY - MOTOR CONTROL
//DESCRIPTION: CONNECT THE RGB LED AND THE POTENTIOMETER TO MAKE THE LED SWITCH COLORS DEPENDING ON THE MOTOR'S SPEED,WHICH
//IS INDICATED BY THE POTENTIOMETER. USE WHILE LOOPS TO CHECK FOR THE POTENTIOMETER'S POSTION
//STARTED: 10/23/18
//FINISHED: 10/25
//DONIYOR NIMATULLO

#include <Servo.h>

Servo myServo;

int potentport = A0, potentvalue, motorpin = 3, red = 7, green = 8,
    blue = 9;

void setup()
{
  myServo.attach(motorpin);
  pinMode(red, OUTPUT);
  pinMode(green, OUTPUT);
  pinMode(blue, OUTPUT);
  Serial.flush();
  Serial.begin(9600);

}

void loop()
{

  potentvalue = analogRead(potentport);
  potentvalue = map(potentvalue, 0, 1023, 0, 180); //maps the value of the potentiometer to the led
  myServo.write(potentvalue); //sets the rotation of the motor
  delay(50);
  RGB();
}

void PrintPotent()
{
  Serial.println("Potentiometer: ");
  Serial.println(potentvalue); //shows the value of the potentiometer
}

void RGB()
{
  while (potentvalue <= 100) //this loop sets the color of the LED to green when the speed value is less than 100
  {
    potentvalue = analogRead(potentport);
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    potentvalue = analogRead(potentport);
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    myServo.write(potentvalue);
    analogWrite(red, 255);
    analogWrite(green, 0);
    analogWrite(blue, 255);
    PrintPotent();
  }
  while (potentvalue >= 101 && potentvalue < 150) //this loop sets the color of the LED to blue when the speed value
  {                                               //is greater than 100 but less than 150
    potentvalue = analogRead(potentport);
    potentvalue = map(potentvalue, 0, 1023, 0, 180);
    myServo.write(potentvalue);
    analogWrite(red, 255);
    analogWrite(green, 255);
    analogWrite(blue, 0);
    PrintPotent();
  }
  while (potentvalue >= 150 && potentvalue <= 180) //this loop sets the color of the LED to red when the speed value
  {                                                //is greater than 150 but less than 180 (which is the max)
    potentvalue = analogRead(potentport);
    potentvalue = map(potentvalue, 0, 1023, 0, 180);
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